to run fast relative to the one at the tail. To see that, imagine that the front clock emits a flash of light each second, and that you are sitting at the tail comparing the arrival of the light flashes with the ticks of clock *B*. Let's say that the rocket is in the position *a* of Figure 6-17 when clock *A* emits a flash, and at the position *b* when the flash arrives at clock *B*. Later on, the ship will be at position *c* when the clock *A* emits its next flash, and at position *d* when you see it arrive at clock *B*.

The first flash travels the distance L_1 and the second flash travels the shorter distance L_2. It is a shorter distance because the ship is accelerating and has a higher speed at the time of the second flash. You can see, then, that if the two flashes were emitted from clock *A* one second apart, they would arrive at clock *B* with a separation somewhat less than one second, since the second flash doesn't spend as much time on the way. The same thing will also happen for all the later flashes. So if you were sitting in the tail you would conclude that clock *A* was running faster than clock *B*. If you were to do the same thing in reverse—letting clock *B* emit light and observing it at clock *A*—you would conclude that *B* was running

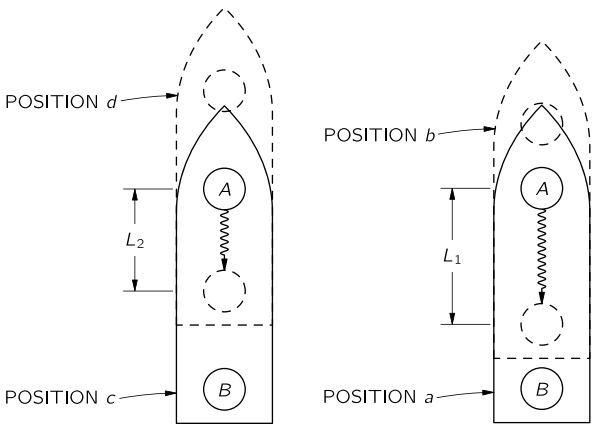


Figure 6-17 A clock at the head of an accelerating rocket ship appears to run faster than a clock at the tail.